

Privacy-Preserving and Scalable Occupant Spatial Tracking in Indoor Environments

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Abstract

Spatiotemporal tracking in indoor environments faces privacy concerns and complexity that affect widespread adoption. This work presents a scalable, privacy-preserving tracking system designed to overcome these barriers.

We propose a wireless, edge-computing architecture that fuses camera and mmWave radar data locally, transmitting only anonymous identity vectors and spatial coordinates. Connectivity is achieved via a self-forming mesh network on the low-interference DECT NR+ (Digital Enhanced Cordless Telecommunications New Radio Plus) in the 1.9 GHz band, ensuring scalable, low-latency communication.

The proposed system will be validated through real-world deployments in Cambridge and London, and will deliver granular tracking data of occupants in buildings while strictly protecting their identities, thereby offering a practical solution for diverse indoor environments.

Introduction

a) Scope and problem definition

Occupant spatial tracking drives critical applications in building optimisation [1] [2], energy reduction [3], shopper behaviour analysis [4][5], and optimising patient flow in clinics to reduce wait times [6] [7]. However, existing solutions have certain privacy risks which prevent people from participating or agreeing to be tracked [6]. In addition, the complex process of installing and deploying the needed infrastructure, such as dedicated wired networks and servers for post-processing, adds further hurdles to practicality and scalability.

This work aims to design an indoor tracking system to deliver spatiotemporal tracking data while being (1) privacy-preserving, (2) scalable, and (3) easy to deploy.

b) State of the art and current limitations

Recent work in [8] [9] focuses on privacy-preserving occupancy detection without retaining images, utilising manually calibrated fields of view (FoV) on Raspberry Pis. While these solutions have made progress in edge-enabled occupancy detection and affordability, they still fail to tackle important scalability challenges: auto-calibration and maximising the number of sensor nodes in the system.

Some existing tracking systems are based on physical tags given to individuals, such as Ubisense [10].

Although such systems usually achieve high accuracy, reaching 30 centimetres in indoor environments, they are considered intrusive, have high privacy implications, and require a complex networking setup. For instance, in a recent NHS trial to track patients, 50% of participants refused to carry a tracking tag [6]. Other systems rely on traditional CCTV footage [11] to produce analytics on individuals seen across multiple cameras by running recognition software on centrally-stored camera recordings. This approach compromises privacy and often lacks the ability to reconstruct complete spatiotemporal journeys.

The closest commercial system to this proposal is Xovis [12], which uses stereo cameras with wired Ethernet to track individuals. However, depending on a wired network for cross-node communication limits scalability and ease of deployment. In addition, this system relies on one type of sensor (stereo cameras), which has certain limitations in poor lighting conditions – a challenge that our proposed radar-camera fusion system directly addresses.

Innovation proposal

a) Differentiating factors of the proposal

To the best of our knowledge, this proposal is the first system that combines radar-camera handoffs with an auto-calibrating, decentralised mesh architecture based on DECT NR+, achieving privacy-preserving and cross-scene tracking.

b) Core concept

We use edge computing to process data directly on the sensor units, ensuring raw sensor data never leaves the device, thereby contributing to preserving privacy. In addition, we use a wireless mesh network architecture to address deployment hurdles.

At the network edge, we fuse camera vision and mmWave radar sensor data to: (1) extract visual features of individuals to create anonymous identity vectors, (2) use the depth data from the privacy-friendly radar to spatial track individuals. The nodes forming the mesh self-coordinate using the unlicensed DECT NR+ 1.9 GHz band, staying away from the crowded 2.4 GHz band [13]. This method offers low interference, enabling longer-range communication and lower latency while providing higher throughput and continuous transmission capabilities [14] [15].

The proposed system delivers granular spatiotemporal data while preserving individual privacy, offering a

solution that is both extendable and easy to deploy across various indoor environments.

The overall system functionality is illustrated in Figure 1. As a person enters the FoV of a sensor node (indicated by 1 on Figure 1), the node calculates a unique ID vector derived from the visual features of the individual (hair colour, worn clothes, gait shape, carried objects), which is then fused with radar’s Doppler and micro-Doppler features [16] (height, distance, speed, gait shape, body part movement patterns). The combination of those features will be used to generate an anonymous signature (feature vector) for the person. As the person moves between overlapping FoVs (2 and 3 on Figure 1), the initial node will hand over the feature vector to the neighbour node to maintain the person’s ID. In the case of a person leaving a FoV and entering a non-overlapping FoV (4 and 5 on Figure 1), the system will transfer the feature vector to the next node, where the person can be re-identified locally and their trajectory reconstructed.

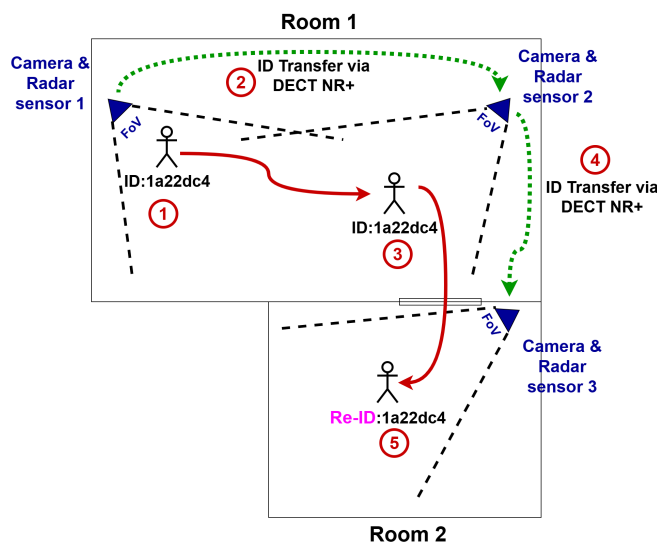


Figure 1: System functionality

The proposed system does not store camera frames or radar data; only the derived features are extracted and transmitted across the off-grid, encrypted DECT NR+ network. Furthermore, the features used to generate the temporary IDs are not stored in the system nodes and are purged once the tracking handover is complete. Ultimately, the system outputs a list of spatial coordinates paired with anonymous IDs, mapping the movement of individuals throughout a building over time.

Vision and plan

This project utilises the Design Science Research framework to design and validate the proposed system (shown in Figure 2).

Figure 3 illustrates the overall plan of the research project.

Currently, in the first year of this project, we focus on testing and selecting the sensors (testing multiple FoV cameras, a different range of mmWave radars). Followed by designing the fusion models on the edge and the calibration procedures for the nodes. By the end of the year, the focus will shift towards designing the mesh network on DECT NR+, including handoff protocols, bandwidth and latency tests.

The second year will focus on the collaboration protocols and the collective behaviour of nodes to achieve the cross-scene tracking capability. The full system assessment will commence afterwards, leading to a thesis submission in the second half of 2029.

In parallel, real-world deployments will be ongoing to gradually validate the system, in addition to developing the Ground Truth methods used to evaluate the system.

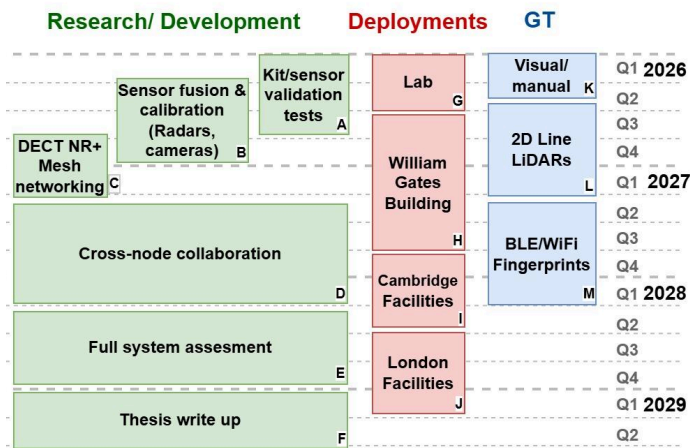


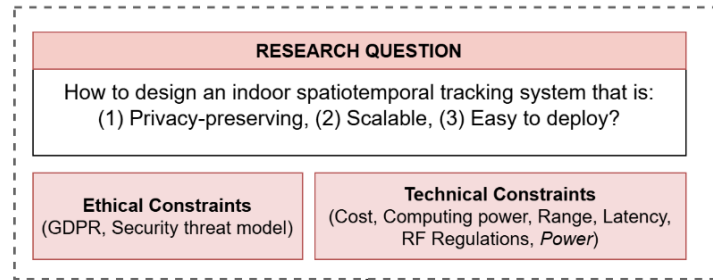
Figure 3: Proposed system timeline (GT: Ground Truth)

Conclusion

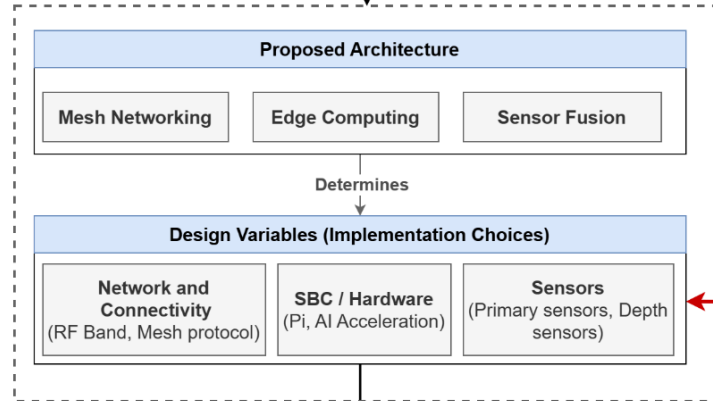
This paper proposes a novel, privacy-preserving, and scalable system for indoor spatiotemporal tracking. By shifting away from centralised networks, we introduce a decentralised edge-computing architecture utilising the low-interference DECT NR+ mesh network. The system utilises radar-camera fusion to track individuals across multiple scenes, generating anonymised trajectories.

Over the next three years, the system will be developed and validated throughout real-world deployments, with preliminary results expected in late 2026, ultimately offering a privacy-preserving, scalable and easy-to-deploy tracking system for the next generation of smart buildings.

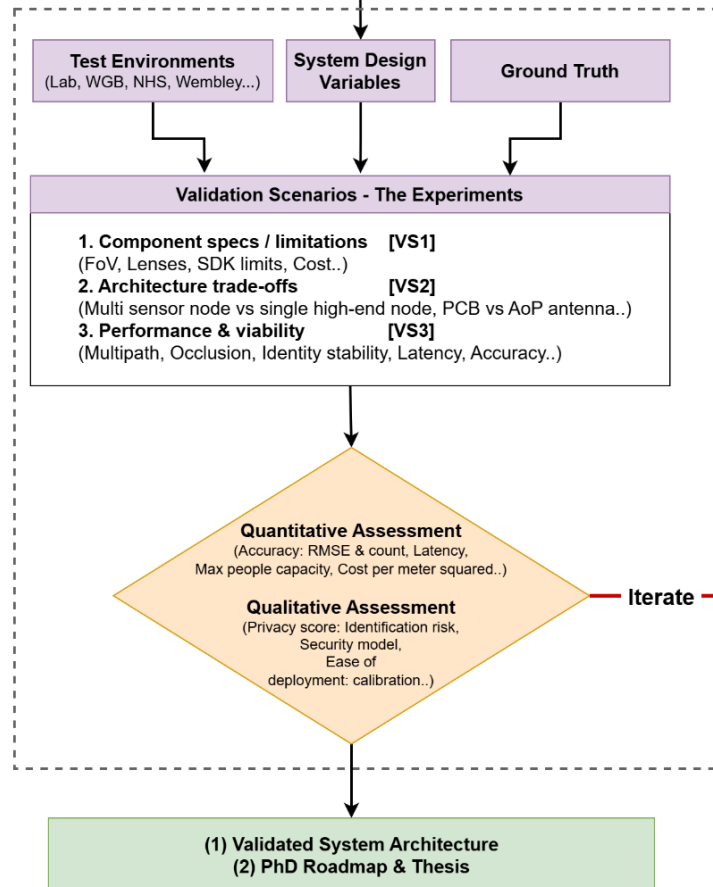
PROBLEM DEFINITION



SYSTEM DESIGN



EXPERIMENTAL VALIDATION



* Based on DSR - Design Science Research framework

Figure 2: The methodology used to design the proposed system – based on DSR framework [17] [18]

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